

QUESTIONS

LEVEL 2 — MEDIUM

Q1

Array

**Q2**

Numbers

**Q3**

Control

Q4

Strings

4 / 4 saved

Level 2 — Medium (M) Matrix Mult. — Array

✓ Submitted

Write a program for matrix multiplication?

Sample Input:

Mat1 =

1 2

5 3

Mat2 =

2 3

4 1

Sample Output:

Mat Sum =

10 5

22 18

Your Code

```
1 import java.util.Scanner;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc= new Scanner(System.in);
5         int[][]a= new int[2][2];
6         int[][]b= new int[2][2];
7         int[][]c= new int[2][2];
8         for(int i=0; i<2; i++)
9             for(int j=0; j<2; j++)
10                 a[i][j]=sc.nextInt();
11         for(int i=0; i<2; i++)
12             for(int j=0; j<2; j++)
13                 b[i][j]=sc.nextInt();
14         for(int i=0; i<2; i++)
15             for(int j=0; j<2; j++)
16                 for(int k=0; k<2; k++)
17                     c[i][j] += a[i][k]*b[k][j];
```

Output

10 5 22 18

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Level 2 — Medium (M) Perfect Num. Gen. — Numbers

✓ Submitted

Write a program to print the first n perfect numbers. (Hint Perfect number means a positive integer that is equal to the sum of its proper divisors)

Sample Input:

N = 3

Sample Output:

First 3 perfect numbers are: 6 , 28 , 496

Testcases:

1. N = 0
2. N = 5
3. N = -2
4. N = -5
5. N = 0.2

Your Code

```
1 class main{
2     public static void main(String[] args){
3         int n = 6;
4         int count=0;
5         int num=1;
6         while(count < 0){
7             int sum =0;
8             for(int i=1; i<num; i++){
9                 if(num%i==0)
10                    sum+=i;
11             }
12             if(sum==num){
13                 System.out.print(num+"");
14                 count++;
15             }
16         }
17     }
18 }
```

Output



QUESTIONS

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Numbers



Q3

Control

Q4

Strings



4 / 4 saved

Level 2 — Medium (M) Palindrome — Strings

✓ Submitted

Write a program using choice to check
Case 1: Given string is palindrome or not
Case 2: Given number is palindrome or not
Sample Input:
Case = 1
String = MADAM
Sample Output:
Palindrome

Testcases:

1. MONEY
2. 5678765
3. MALAY12321ALAM
4. MALAYALAM
5. 1234.4321

Your Code

```
1 import java.util.*;  
2 class main{  
3     public static void main(String[] args){  
4         Scanner sc = new Scanner(System.in);  
5         String s = sc.next();  
6         int n = sc.nextInt();  
7  
8         String rev = "";  
9         for(int i=s.length()-1; i>=0; i--)  
10             rev+=s.charAt(i);  
11         if(s.equals(rev))  
12             System.out.println("String is palindrome ");  
13         else  
14             System.out.println("String is not a palindrome");  
15     }  
16 }
```

Output

```
Error: Execution timeout (>5 seconds) -  
possible infinite loop
```

QUESTIONS

LEVEL 2 — MEDIUM

Q1

Array



Q2

Numbers



Q3

Control

Q4

Strings



4 / 4 saved

Level 2 — Medium (M) Tax — Control

Write a program to calculate tax given the following conditions:

- a. If income is less than or equal to 1,50,000 then no tax
- b. If taxable income is 1,50,001 – 3,00,000 the charge 10% tax
- c. If taxable income is 3,00,001 – 5,00,000 the charge 20% tax
- d. If taxable income is above 5,00,001 then charge 30% tax

Sample Input:

Enter the income:200000

Sample Output:

Tax= 20000

Testcases:

Test cases:

- 1. 400700
- 2. 2789239
- 3. 150000
- 4. 00000
- 5. -125486

Your Code

```
1 import java.util.Scanner
```

Program Input (stdin)

Enter input if required...

Output